

The Ephemeris

March 2019

Volume 30 Number 01- The Official Publication of the San Jose Astronomical Association



March 2019 - June 2019 Events

Board & General Meetings

Saturday 3/23, 4/20, 5/18, 6/15

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 3/3, 4/7, 5/5, 6/2

Solar Sunday (1:30-3:30pm)

Sunday 3/3, 4/7, 5/5, 6/2

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 3/15, 4/12, 5/10, 6/7

Houge Park 3Q In-Town Star Party

Friday 3/29, 4/26, 5/24, 6/21

RCDO Starry Nights Star Party

Saturday 3/30, 4/27, 5/25, 6/22

Imaging SIG Mtg

Tuesday 3/19, 4/16, 5/21, 6/18

Astro Imaging Clinics (at Coyote Valley OSP)

Saturday 3/9, 5/4, 6/29

Quick START

Quarterly (by appointment)

Binocular Star Gazing

Will resume in Summer 2019

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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Photo by SJAA Member: Everett Quebral
October 14, 2018
Location: San Jose, CA

The Wizard Nebula
backyard-astronomer.com

SJAA Awards and Board Elections 2019

By Vikas Kache

On Saturday February 16th, 2019 the SJAA board of directors had a meeting and a potluck dinner. The night kicked off with a board of directors meeting, followed by an awards ceremony and then dinner with elections.

SJAA's annual recognition awards are given to those individuals who have provided exemplary service to SJAA, typically going above and beyond their assigned duties. This year, SJAA is recognizing three such individuals who have not only been long time contributors to SJAA but have stepped up to make a significant improvement to the programs they lead. The awards were presented by the president of the SJAA, Swami Nigam.



SJAA President Swami Nigam presents an award to Vini Carter

Vini Carter has been running the Fixit program, and also managing the SJAA sound system during our general meetings. The Fixit program provides free help to club members as well as public in fixing simple problems with their telescopes and other astronomical gear, as well as collimation and general help in operating and maintaining a telescope. For most of last year, Vini has been running the program single handedly. There isn't a telescope problem that Vini can't fix!



SJAA President Swami Nigam presents an award to Muditha Kanchana

Muditha Kanchana, better known as Kanch, is the leader of the loaner program for the last few years. The loaner program loans out telescopes for a short duration to SJAA members who may not currently own their own telescope and would like to try one out before investing into a gear of their own. Kanch has made significant improvement to this program by increasing our telescope inventory and getting the program better organized. Kanch is also currently working on making our storage space better organized and secure.



SJAA President Swami Nigam presents an award to Carl Wong

Carl Wong took over the very popular Starry Nights Star Party program that SJAA conducts in collaboration with Santa Clara Valley OSA to provide a dark sky observing experience to the general public every month. Carl stepped up to fill this much required role, and has been working on creating a streamlined process for SJAA astronomers to volunteer for this monthly events, as well as training newbie astronomers on how to engage with the public during such events. Working in partnership with the OSA, as well as getting stakeholders involved, Carl is helping to make the program thrive.

What we hope is for more SJAA members to get more involved in these efforts, and maybe you can be recognized next time!

In addition, a special shout-out went to Marianne Damon, Sukhada Palav, and Wolf Witt, for the tremendous work they have been doing in their departments year after year.

For the elections, Vini Carter, Rob Chapman, Ken Miura, Swami Nigam, and Sukhada Palav were re-elected into their positions. These elections were for the Director position on the Board of Directors. Elections were followed by a delicious potluck with food and treats brought by SJAA members.

Tour Alien world with new multimedia treats

Credit: NASA JPL



This Exoplanet Travel Bureau poster illustration shows futuristic explorers gliding in a protective bubble over the red-hot landscape of the exoplanet 55 Cancri e. Exoplanets are planets outside our solar system.

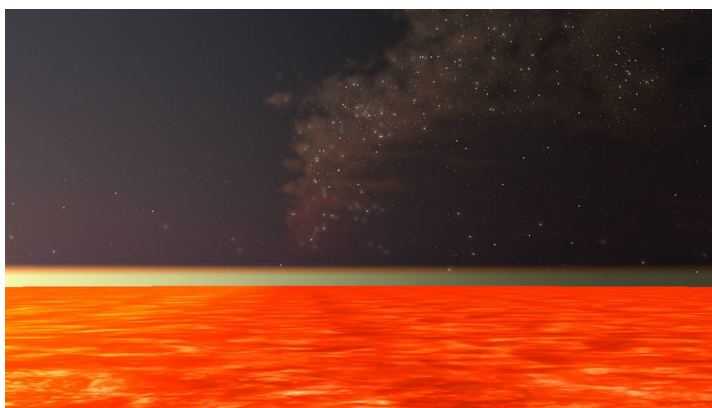
Credits: NASA/JPL-Caltech

Explore the plethora of planets outside our solar system with new multimedia experiences from NASA's Exoplanet Exploration Program (ExEP). In addition to a new Exoplanet Travel Bureau poster celebrating a molten world called 55 Cancri e, space fans can enjoy a 360-degree visualization of the surface of the same planet, a multimedia journey into the life and death of planetary systems, and a major update to the popular Eyes on Exoplanets app.

Lava Life

Designed in the style of vintage travel posters, ExEP's popular Exoplanet Travel Bureau poster series imagines what it might be like to visit known planets outside our solar system, or exoplanets. Focusing on 55 Cancri e, a planet that may be covered in a lava ocean, the newest poster shows futuristic explorers gliding over the red-hot landscape in a protective bubble.

55 Cancri e is also now part of the Exoplanet Travel Bureau's 360-degree visualization tool, which enables you to take a virtual tour of what the planet's surface might look like, based on the limited data available (no photos of the



This artist's illustration from the Exoplanet Travel Bureau's 360-degree visualization tool reveals what the surface of exoplanet 55 Cancri e might look like, based on the limited data available. This exoplanet (a planet outside our solar system) is thought to be covered entirely in molten lava.

Credits: NASA/JPL-Caltech

planet exist). Seen as a massive fiery orb on the horizon, the planet's star is 65 times closer to 55 Cancri e than the Sun is to Earth. On the planet's cooler nightside, silicate vapor in the atmosphere may condense into sparkling clouds that reflect the lava below.

All of the 360-degree visualizations are viewable on desktop computers, mobile devices and through virtual reality headsets that work with smartphones.

Life and Death of a Solar System

How did we get here? How do stars and planets come into being, and what fate awaits planets after their stars die? The interactive web feature "Life and Death of a Planetary System" brings readers on an in-depth journey through the formation, evolution and eventual demise of a solar system. This multichapter story offers insight into how the planet we call home formed and what will happen to it when the Sun dies.

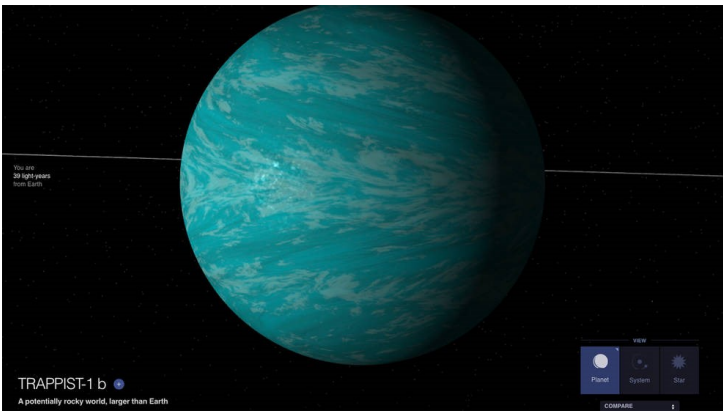
Planet Bonanza

Explore thousands of new worlds, both strange and strangely familiar, with NASA's Eyes on Exoplanets 2.0. Users can fly through the galaxy and virtually visit any of the nearly 4,000 known exoplanets, all visualized in 3-D. Interstellar ports of call include the TRAPPIST-1 system of seven Earth-sized planets, the potentially molten-lava-covered 55 Cancri e, the egg-shaped WASP-12b and Kepler-16b, the first world discovered orbiting two stars.

(Continued on the following page)

(Tour Alien—continued from page 3)

Among other features, the searchable Eyes on Exoplanets 2.0 lets users compare an exoplanet's size to that of Earth or Jupiter; determine how long it would take to travel to a given planet by car, jet or light-speed starship; and interact with virtual models of NASA space telescopes, such as Hubble, Spitzer, Kepler and the newly launched Transiting Exoplanet Survey Satellite (TESS).



This artist's illustration of a planet in the TRAPPIST-1 system can be found in NASA's Eyes on Exoplanets 2.0. The web-based program lets users virtually fly through the galaxy and visit any of the nearly 4,000 known exoplanets, all visualized in 3-D.

Credits: NASA/JPL-Caltech

Eyes on Exoplanets 2.0 is powered by data from NASA's Exoplanet Archive, the official database used by scientists researching exoplanets. Available for use on desktop computers as well as most smartphones and tablets, this next-generation, browser-based version of the popular app requires no software download.

The Exoplanet Travel Bureau was developed by NASA's Exoplanet Exploration Program communications team and program chief scientists. Based at the agency's Jet Propulsion Laboratory in Pasadena, California, which is a division of Caltech, the program leads NASA's search for habitable planets and life beyond our solar system. The program develops technology and mission concepts, maintains exoplanet data archives and conducts ground-based exoplanet science for NASA missions.



All Systems Go As Parker Solar Probe Begins Second Sun Orbit

Credit: NASA

On Jan. 19, 2019, just 161 days after its launch from Cape Canaveral Air Force Station in Florida, NASA's Parker Solar Probe completed its first orbit of the Sun, reaching the point in its orbit farthest from our star, called aphelion. The spacecraft has now begun the second of 24 planned orbits, on track for its second perihelion, or closest approach to the Sun, on April 4, 2019.

Parker Solar Probe entered full operational status (known as Phase E) on Jan. 1, with all systems online and operating as designed. The spacecraft has been delivering data from its instruments to Earth via the Deep Space Network, and to date more than 17 gigabits of science data has been downloaded. The full dataset from the first orbit will be downloaded by April.

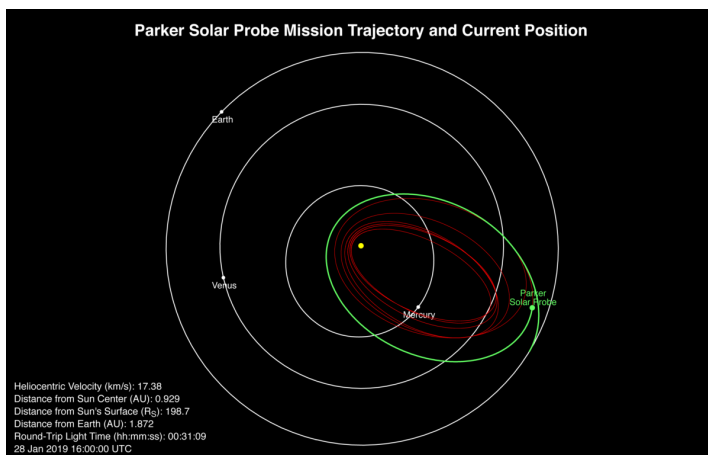
"It's been an illuminating and fascinating first orbit," said Parker Solar Probe Project Manager Andy Driesman, of the Johns Hopkins University Applied Physics Laboratory. "We've learned a lot about how the spacecraft operates and reacts to the solar environment, and I'm proud to say the team's projections have been very accurate." APL designed, built, and manages the mission for NASA.

"We've always said that we don't know what to expect until we look at the data," said Project Scientist Nour Raouafi, also of APL. "The data we have received hints at many new things that we've not seen before and at potential new discoveries. Parker Solar Probe is delivering on the mission's promise of revealing the mysteries of our Sun."

In preparation for that next encounter, the spacecraft's solid state recorder is being emptied of files that have already been delivered to Earth. In addition, the spacecraft is receiving updated positional and navigation information (called ephemeris) and is being loaded with a new automated command sequence, which contains about one month's worth of instructions.

Like the mission's first perihelion in November 2018, Parker Solar Probe's second perihelion in April will bring the spacecraft to a distance of about 15 million miles from the Sun – just over half the previous close solar approach record of about 27 million miles set by Helios 2 in 1976.

The spacecraft's four instrument suites will help scientists begin to answer outstanding questions about the Sun's fundamental physics — including how particles and solar material are accelerated out into space at such high speeds and why the Sun's atmosphere, the corona, is so much hotter than the surface below.



Parker Solar Probe's position, speed and round-trip light time as of Jan. 28, 2019. Track the spacecraft online.

The Parker Solar Probe team is not only focused on analyzing the science data but also preparing for the second solar encounter, which will take place in about two months.



Date: Jan 23 2019

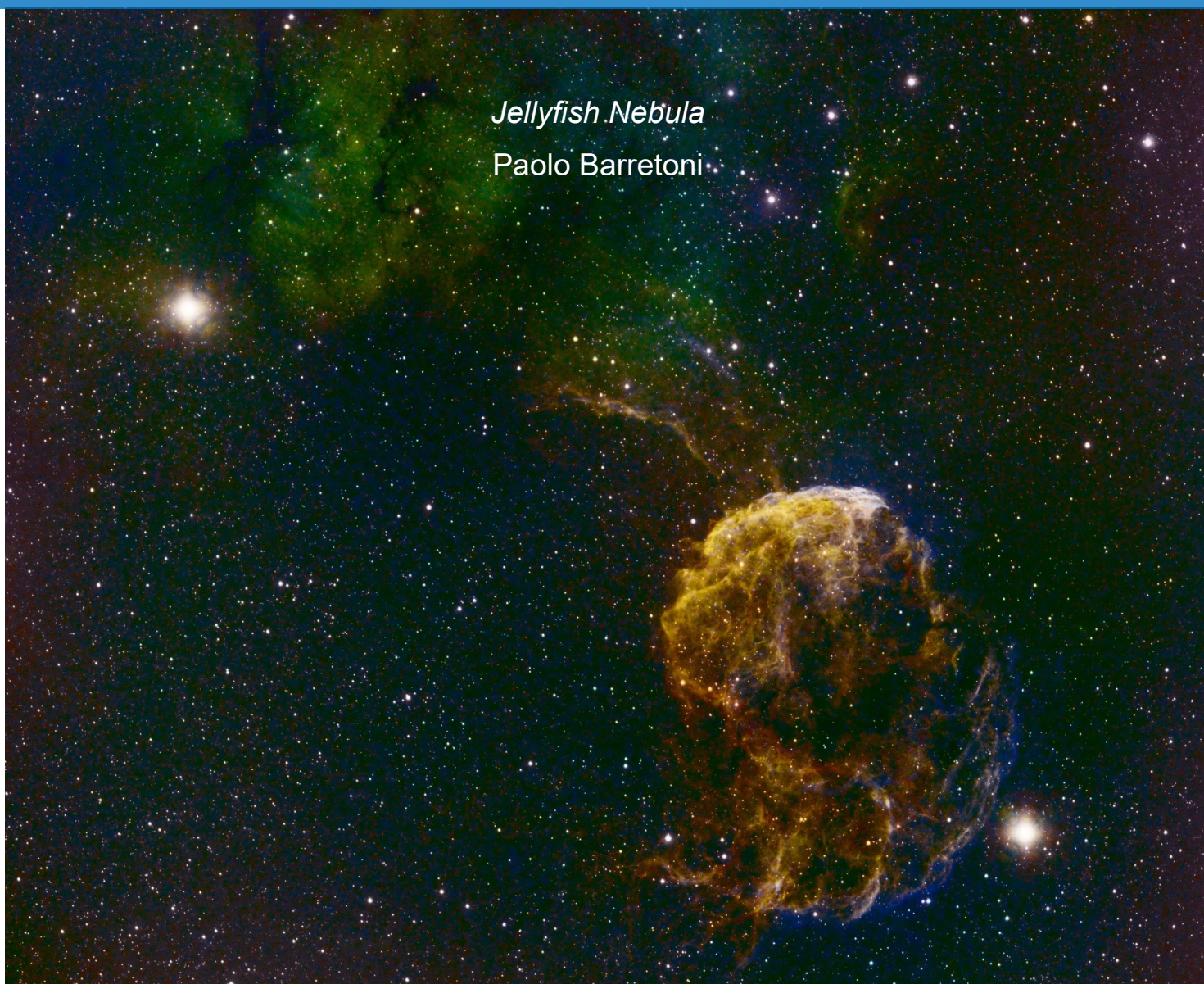
Location: Mountain View, CA

Telescope: Stellarview 9x50 Guide Scope

Camera: Nikon D5300

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight 1.8 Ripely



Dates: Jan 21, 2019, Jan 22, 2019, Jan 23, 2019

Location: San Jose, CA

Telescope: Canon 400mm

Camera: Orion Star Shoot AutoGuider (SSAG)

Mount: Orion Atlas AZ/EQ Pro

Software: Main Sequence Software Sequence Generator Pro 3, PixInsight, PHD2

SJAA Member Astrophoto Gallery



Wide field view towards star-forming nebula
NGC 3603

Nikola Nikolov

NGC 3603 Star-Forming Nebula
Nikola Nikolov

Date: Feb 16 2019

Location: Siding Springs observatory, Coonabarabran, New South Wales, Australia (I Telescope)

Telescope: Takashi ED 106 320mm

Camera: FLI Microline 16803

Mount: Paramount GTS

Software: Photoshop CC 2018 Photoshop CC2018, AstroPixelProcessor

SJAA Member Astrophoto Gallery



Date: Nov 7 2018

Location: Stellar Skies Observatory, Pontotoc, Tx

Telescope: Orion 10" f/3.9 Astrograph

Camera: QSI 683 wsg-8

Mount: Software Bisque Paramount MYT

Software: Main Sequence Software Sequence Generator Pro, Software Bisque Sky X Pro, PHD2 Guiding, Pix-Insight 1.8 PixInsight

Kid Spot



Kid Spot Jokes:

- ♦ **What is an insect's favorite star?**
Betel-geuse.
- ♦ **What is an alien's favorite candy?**
Milky Way
- ♦ **What does a body builder in space take?**
A-steroids

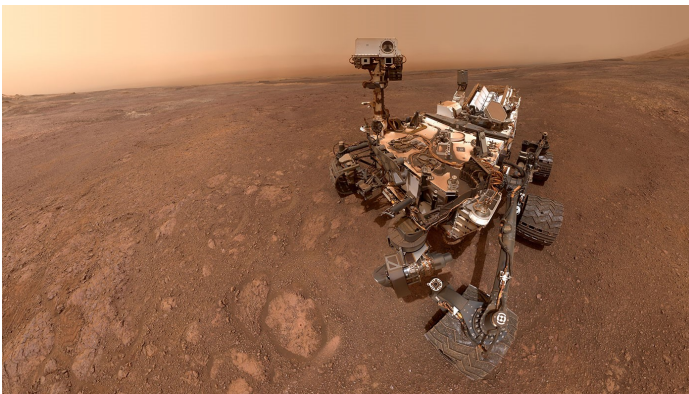
Kid Spot Quiz:

1. **The Pillars of Creation as seen in the Hubble Space Telescope is part of which nebula?**
a) Trifid Nebula b) Orion Nebula c) Eagle Nebula
2. **In which constellation is the double star Alberio?**
a) Cygnus b) Gemini c) Taurus
3. **By what colloquial name is Pleiades Cluster, M45 also known?**
a) Seven Princes b) Seven Sisters c) Shepherd's Bush



Curiosity Says Farewell to Mars' Vera Rubin Ridge

Credit: NASA JPL



A selfie taken by NASA's Curiosity Mars rover on Sol 2291 (January 15) at the "Rock Hall" drill site, located on Vera Rubin Ridge. Image Credit: NASA/JPL-Caltech

On Dec. 15, Curiosity drilled its 19th sample at a location on the ridge called Rock Hall. On Jan. 15, the spacecraft used its Mars Hand Lens Imager (MAHLI) camera on the end of its robotic arm to take a series of 57 pictures, which were stitched together into this selfie. The "Rock Hall" drill hole is visible to the lower left of the rover; the scene is dustier than usual at this time of year due to a regional dust storm.

Curiosity has been exploring the ridge since September of 2017. It's now headed into the "clay-bearing unit," which sits in a trough just south of the ridge. Clay minerals in this unit may hold more clues about the ancient lakes that helped form the lower levels on Mount Sharp.

NASA's Curiosity rover has taken its last selfie on Vera Rubin Ridge and descended toward a clay region of Mount Sharp. The twisting ridge on Mars has been the rover's home for more than a year, providing scientists with new samples - and new questions - to puzzle over.

From the Board of Directors

Announcements

- ◇ Ken Miura has chosen to take over the Stars Over Yosemite position
- ◇ Vikas Kache has been elected as the new Ephemeris Content Editor
- ◇ Kaushal Gangakhedkar has been elected as the new Membership Chair position
- ◇ SJAA Spring Swap Meet is scheduled for March 17, 2019, 12 noon.
- ◇ Officer Elections will be during the March board meeting scheduled for March 23, 2019.

Board Meeting Excerpts

November 17, 2018

In attendance

Swami, Rob J, Sukhada, Glenn, Ken, Vini, Rob C, Sandy
Excused Absences: Wolf
Guest: Marianne, Tracy

A lot of open positions

SJAA is trying to fill many open positions: Vice President, Ephemeris Content Editor, Membership Chair, Donations Chair, Admin for the Observers Yahoo Group, Meetup, etc.

Yosemite Leader

Rob Jaworski took over trying to get in touch with the National Park Services. Ken Miura volunteered to help with the program.

Lunar Eclipse on Jan 20

There will be a lunar eclipse on Jan 20. OSA would like to hold a joint event with SJAA at Little Uvas Creek Park. However, you don't need a dark site for lunar eclipse. So, SJAA will advise OSA to hold the event at Hogue Park.

RASC Handbooks and Calendars

It's time to order RASC handbooks and calendars for 2019. Since the handbooks are readily available in many

places, SJAA will only order calendars this year.

Swap Meet recap

Swap Meet was well attended and went smoothly. Next swap meet is scheduled for 16th of March

Walden West

Tracy Avent-Costanza has been working with Walden West to support them with their astronomy night. Tracy is relocating, so will not be able to continue supporting them. He is looking for someone to take over.

December 15, 2018

In attendance

Sandy, Wolf, Vini, Rob J, Glenn, Ken
Excused Absences: Rob C, Swami, Sukhada
Guest: Steven, Dave

Open positions

Open positions are still open. No progress so far.

2019 Events Calendar

2019 SJAA events calendar will be completed by the end of the year.

Board Elections

Board elections are coming up in February. 5 of the 9 board positions will be up for elections.

Astronomy League

It has been proposed that SJAA become an affiliate of Astronomy League. Steven Green has volunteered to research this.

January 19, 2019

In attendance

Swami, Rob J, Wolf, Vini, Sukhada, Glenn, Sandy
Excused Absences: Ken
Guest: Kaia, Mo

Ephemeris Content Editor

Vikas Kache is the new Ephemeris Content Editor, taking over from Sandy Mohan.

Membership Chair position

Many responses to Dave's message on Meetup about open Membership Chair position.

Upcoming Board Elections

Rob Chapman will send out an announcement about the upcoming board elections to SJAA members

Nominations for annual awards

Nominations are invited for the annual SJAA recognition awards.

Heidi Gerster donation

Heidi Gerster made a generous donation of \$500 to be added to the Armstrong scholarship

Get more members involved

Swami made a short presentation on how to increase membership participation in club operations, and some suggestions for 2019. Several new positions have been proposed, which we will start filling soon.

February 16, 2019

In attendance

Swami, Rob C, Sukhada, Glenn, Ken Miura, Vini, Wolf
Excused Absences: Sandy, Rob J
Guests: Emi, Kaia C, Dave, Ed, Vikas, Marianne, Carl, Kanch, Jim

New Membership Chair

Kaushal Gangakhedkar is the new Membership Chair, taking over from Dave Ittner.

Stars Over Yosemite program

We have received information from Retired Ranger David Balogh over Stars Over Yosemite program. Ken will review and submit SJAA application.

Kelley Park

Kelley Park rangers are requesting SJAA participation in some of their events.

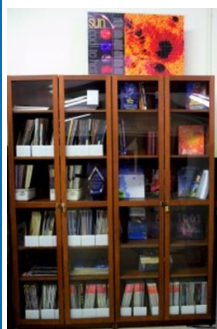
Scout-O-Rama Boy scout event

There is a Scout-O-Rama Boy scout event coming up on May 11, and they have invited SJAA to setup a booth. Feasibility and details are being worked out.

Officer Elections

Officer elections for President, Vice-President, Treasurer, and Secretary positions will be held during the March board meeting. Any SJAA member in good standing that is interested in running for an officer seat is welcome to come to the board meeting on 23 March 2019. Please see the organization's by-laws for details.
<https://www.sjaa.net/about/club-by-laws/#A4>

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STArT Program

The Quick STArT Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content chang-

ing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick STArT program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Swami Nigam
Vice President:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Sandy Mohan
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Vikas Kache
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Quick STArT	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with **online** Ephemeris

☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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Address:

City/ST/Zip:

Phone:

E-mail address:
